Parallel Physics-Informed Prognostics for EV Battery Health Management: Fusing Physics Constraints, Uncertainty and GPU-Scale Inference

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THE BREAKTHROUGH

To our knowledge, **the first framework to combine physics-informed learning, calibrated RUL distributions and GPU-scale particle inference** — the four capabilities every prior approach leaves incomplete — validated on real NASA PCoE cells across **40 measured configurations on two machines**.

RESEARCH GAP — WHAT NO ONE HAS COMBINED 01

Battery RUL research has four mature ingredients. No published framework combines them — a maintenance decision needs all four at once.

approach family	physics informed	RUL distribution	GPU scale	real-cell benchmark
Physics-informed NNs [13], [22]	●	⊗	⊗	partial
LSTM + particle filter [27], [28]	⊗	●	⊗	single cell
CUDA-parallel particle filters [17]	⊗	●	●	not battery
P3-EVHM — this work	●	●	●	●

Verified against a systematic review of 33 works: the closest LSTM+filter hybrid [28] explicitly lists our physics constraints as *future work*; CUDA particle filters [17] never touched battery data. **P3-EVHM closes all four cells — that closure is the contribution.**

THREE CONTRIBUTIONS 02

C1 · PHYSICS

~5×

fewer impossible predictions

Two electrochemistry laws enter the loss (Eq. 4–5): capacity never rises; decline respects an Arrhenius temperature prior. Violations 0.366 → 0.076; capacity MAE improves 0.285 → 0.257 Ah.

C2 · UNCERTAINTY

20/20

configs give a calibrated RUL

Thousands of simulated futures reweighted by the measured capacity (Eq. 6–8): median + 95% interval + decision rules from its lower bound. Mean error 18.8 cycles, coverage 0.85, on both machines.

C3 · GPU SCALE

16.5×

at 500k particles, open-sourced

One vectorised tensor op per step (SIMT), 20.2 ms device time. MIT framework: pip package, dashboard, CI, Kaggle kernel — 40 measured configs as evidence.

METHOD — THE PHYSICS ENTERS THE LOSS 03

(4) $L = MSE(\hat{q}, q) + \lambda_1 \cdot mean\ ReLU(\hat{q}_t - q_t) + \lambda_2 \cdot mean(\Delta\hat{q}_t - g(T_t))^2$

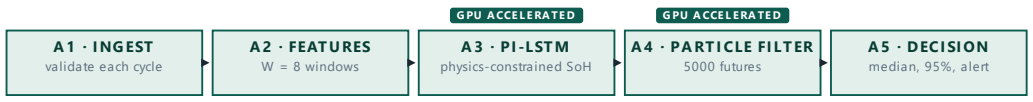
(5) $g(T) = clamp(\hat{A} \cdot (T - \bar{T}), min = 0)$ — *prior fitted from training cycles, not a hard clamp*

(6) $q_i^{(k)} = q_i^{(k-1)} - d_i + \varepsilon_k,\ \ w_i \propto exp(-(q_{obs} - q_i)^2 / 2\sigma^2)$

(8) $RUL = median\{\, k : q_i^{(k)} \le q_{EOL} \},\ \ report\ [P_{2.5}, P_{97.5}]$

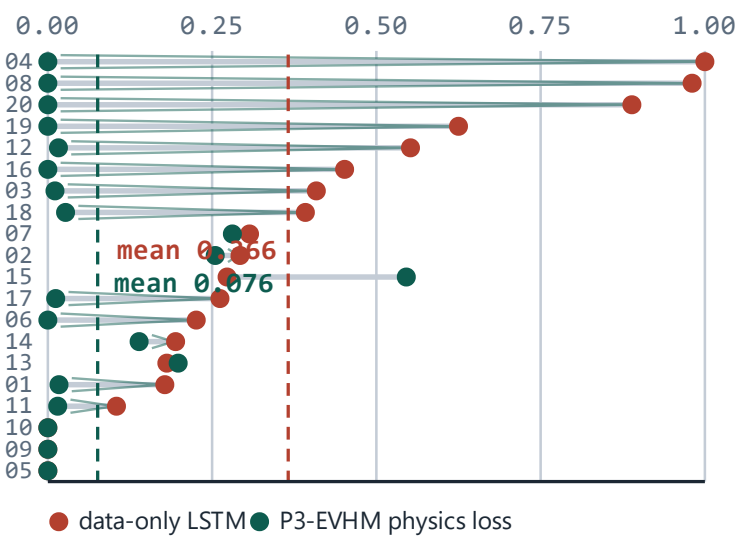
A 1–32 LSTM over a W = 8 window; identical seeds and metrics for baselines B1–B6, so the comparison is the physics, not the tuning. Same code runs on CPU or CUDA (`device="auto"`).

SYSTEM — FIVE AGENTS, ONE FRAMEWORK 04



A3 and A4 run on the GPU; the same code runs on CPU for parity. Four doors into one engine: Python API, CLI, dashboard, Kaggle kernel.

EVIDENCE FOR C1 — THE PHYSICS LOSS WORKS 05



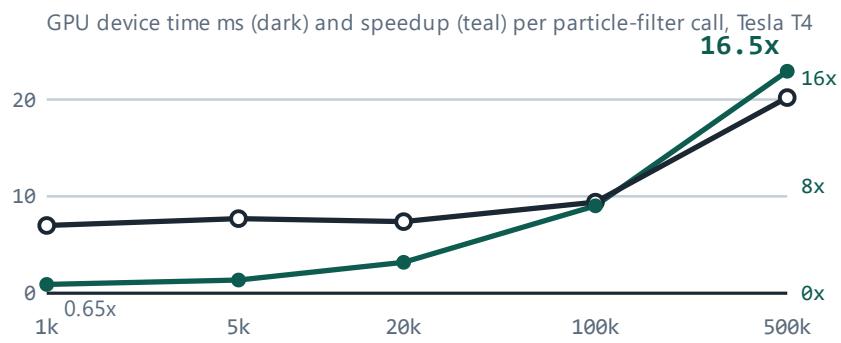
Violation rate V per configuration (share of predicted steps where capacity *rises* — physically impossible). All 20 configurations improve or tie; the two catastrophes (V = 1.0, configs 04/08) are eliminated entirely.

EVIDENCE FOR C2 — PROGNOSTIC QUALITY 06

QUANTITY	MEASURED	NOTE
RUL produced	**20 / 20**	every config, CPU *and* T4
mean RUL error	**18.8 cyc**	18.6 on CPU — reproducible
95% interval coverage	**0.85**	honest, below the 0.95 ideal
transfer: held-out cells	**4 / 4**	vs 1/4 from scratch; 0.2 s head-only adaptation

A 21 KB pretrained checkpoint, body frozen, head fine-tuned: RUL in **4/4 held-out cells** vs **1/4** from scratch; zero physics violations vs 0.355 max. Transfer is what a fleet needs — no per-cell training.

EVIDENCE FOR C3 — GPU SCALING 07



Honest crossover reported: below ~5k particles the warm CPU wins (0.65×); the GPU pays exactly where prognostics needs it — large clouds, sharp intervals, real time.

CONCLUSIONS & FUTURE WORK 08

Conclusions. The fusion is the contribution: physics constraints make LSTM prognostics physically consistent at no accuracy cost; the particle filter turns that forecast into a calibrated maintenance decision in every configuration tested; GPU-scale inference makes the cloud size that sharp intervals need affordable. Open-sourced end to end, with 40 measured configurations as evidence.

Future work. Pack-level validation beyond single cells; prior-floor tuning for early-start predictions (40% of life); interval coverage toward 0.95; streaming on-vehicle deployment.

Code & full validation: github.com/Jithendra123892/p3-evhm (MIT) · GPU kernel: jithendramaddala/p3-evhm-gpu-run **Reproduce:** pip install -e . · python -m p3evhm.demo · streamlit run app.py

Key references [1] Saha & Goebel, *Battery Data Set*, NASA PCoE, 2007. [2] Arulampalam et al., *IEEE Trans. Signal Process.* 50(2), 2002. [13] Shi, Rivera & Wu, *Mech. Syst. Signal Process.* 179, 2022. [17] *Parallel accelerated CRPF on CUDA*, Appl. Intell., 2020. [22] do Nascimento et al., *Sci. Rep.* 13, 2023. [27] *AI-driven particle filter for battery RUL*, Batteries 10(12), 2024. [28] *Hybrid particle-filter-trained neural network*, IEEE, 2021. — full 33-entry bibliography in the paper; all figures generated from measured results.